SOLVING SYSTEMS OF LINEAR EQUATIONS BY MATRIX

LINEAR ALGEBRA

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Solving System of Linear Equations by Matrix

SOLVING SYSTEM OF LINEAR EQUATIONS BY MATRIX

Solve the system of linear equations using **matrix**

$$a - 2b = -10$$

$$3a + 5b = 14$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{ccc} a & b & \end{array} \\ \begin{array}{c} -3 \\ \\ \end{array} & \begin{bmatrix} -3 & 6 & 30 \\ 1 & -2 & -10 \\ 3 & 5 & 14 \end{bmatrix} \end{array} \begin{array}{l} \\ r_1 \\ r_2 \to -3r_1 + r_2 \end{array}$$

$$\begin{array}{c} \begin{array}{ccc} a & b & \end{array} \\ \begin{bmatrix} 1 & -2 & -10 \\ 0 & 11 & 44 \end{bmatrix} \end{array} \begin{array}{l} r_1 \\ r_2 \end{array}$$

Solve the system of linear equations using **matrix**

$$a - 2b = -10$$

$$3a + 5b = 14$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{ccc} a & b & \end{array} \\ & \begin{bmatrix} 1 & -2 & -10 \\ 0 & 11 & 44 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

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Augmented matrix,

$$\begin{array}{cc} & \begin{array}{ccc} a & b & \end{array} \\ & \begin{bmatrix} 1 & -2 & -10 \\ 0 & 11 & 44 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

$$0a + 11b = 44$$

Solve the system of linear equations using **matrix**

$$a - 2b = -10$$

$$3a + 5b = 14$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{ccc} a & b & \end{array} \\ & \begin{bmatrix} 1 & -2 & -10 \\ 0 & 11 & 44 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

Solution,

$$\cancel{0a} + \frac{11b = 44}{\cancel{11}}$$

$$b = 4$$

$$\boxed{b = 4}$$

ans

Solve the system of linear equations using **matrix**

$$a - 2b = -10$$

$$3a + 5b = 14$$

Solution,

$$\boxed{b = 4}$$

ans

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{ccc} a & b & \end{array} \\ & \begin{bmatrix} 1 & -2 & -10 \\ 0 & 11 & 44 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

$$a - 2b = -10$$

Solve the system of linear equations using **matrix**

$$a - 2b = -10$$

$$3a + 5b = 14$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{ccc} a & b & \end{array} \\ & \begin{bmatrix} 1 & -2 & -10 \\ 0 & 11 & 44 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

Solution,

$$a - 2b = -10$$

$$a - 2(4) = -10$$

$$\boxed{b = 4}$$

ans

Solve the system of linear equations using **matrix**

$$a - 2b = -10$$

$$3a + 5b = 14$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{ccc} a & b & \end{array} \\ & \begin{bmatrix} 1 & -2 & -10 \\ 0 & 11 & 44 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

Solution,

$$a - 2b = -10$$

$$a - 8 = -10$$

$$\boxed{b = 4}$$

ans

Solve the system of linear equations using **matrix**

$$a - 2b = -10$$

$$3a + 5b = 14$$

Solution,

$$\boxed{b = 4}$$

ans

$$\boxed{a = -2}$$

ans

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{ccc} a & b & \end{array} \\ & \begin{bmatrix} 1 & -2 & -10 \\ 0 & 11 & 44 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \end{array}$$

$$a - 2b = -10$$

$$a = -10 + 8$$

$$a = -2$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ -1 & 2 & -1 & 0 \\ 5 & -5 & 3 & 6 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

echelon form

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ -2 & 4 & -2 & \\ -1 & 2 & -1 & 0 \\ 5 & -5 & 3 & 6 \end{bmatrix} \begin{array}{l} r_1 \\ \\ r_2 \rightarrow 2r_2 + r_1 \\ r_3 \end{array} \\ 2 & \end{array}$$

$$\begin{array}{c} \begin{array}{cccc} a & b & c & \end{array} \\ \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & 0 & -4 & 12 \\ 5 & -5 & 3 & 6 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

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Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{c} \\ 5 \\ \\ -2 \\ \\ \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & \end{array} \\ \begin{bmatrix} 10 & -20 & -10 & 60 \\ 2 & -4 & -2 & 12 \\ -10 & 10 & -6 & -12 \\ 5 & -5 & 3 & 6 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{l} \\ r_1 \\ \\ r_2 \rightarrow 5r_1 + -2r_2 \\ r_3 \end{array}$$

$$\begin{array}{c} \begin{array}{ccc} a & b & c \end{array} \\ \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{c} \begin{array}{cccc} a & b & c & \end{array} \\ \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

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Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solution,

$$\boxed{c = -3}$$

ans

Augmented matrix,

$$\begin{array}{cccc} a & b & c & \\ \end{array}$$

$$\begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}$$

$$\frac{-4c = 12}{-4}$$

$$c = -3$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

$$-10b - 16c = 48$$

Solution,

$$\boxed{c = -3}$$

ans

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

$$-10b - 16c = 48$$

$$-10b - 16(-3) = 48$$

Solution,

$$\boxed{c = -3}$$

ans

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

$$-10b - 16c = 48$$

$$-10b + 48 = 48$$

Solution,

$$\boxed{c = -3}$$

ans

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

Solution,

$$-10b - 16c = 48$$

$$-10b = 48 - 48$$

$$\boxed{c = -3}$$

ans

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solution,

$$\boxed{c = -3}$$

ans

$$\boxed{b = 0}$$

ans

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

$$-10b - 16c = 48$$

$$-10b = 0$$

$$b = 0$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$2a - 4b - 2c = 12$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$2a - 4b - 2c = 12$$

$$2a - 4(0) - 2(-3) = 12$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$2a - 4b - 2c = 12$$

$$2a \qquad - 2(-3) = 12$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$2a - 4b - 2c = 12$$

$$2a \qquad + 6 = 12$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$2a - 4b - 2c = 12$$

$$2a \qquad = 12 - 6$$

Solve the system of linear equations using **matrix**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

Solution,

$\boxed{c = -3}$ *ans*

$\boxed{b = 0}$ *ans*

$\boxed{a = 3}$ *ans*

Augmented matrix,

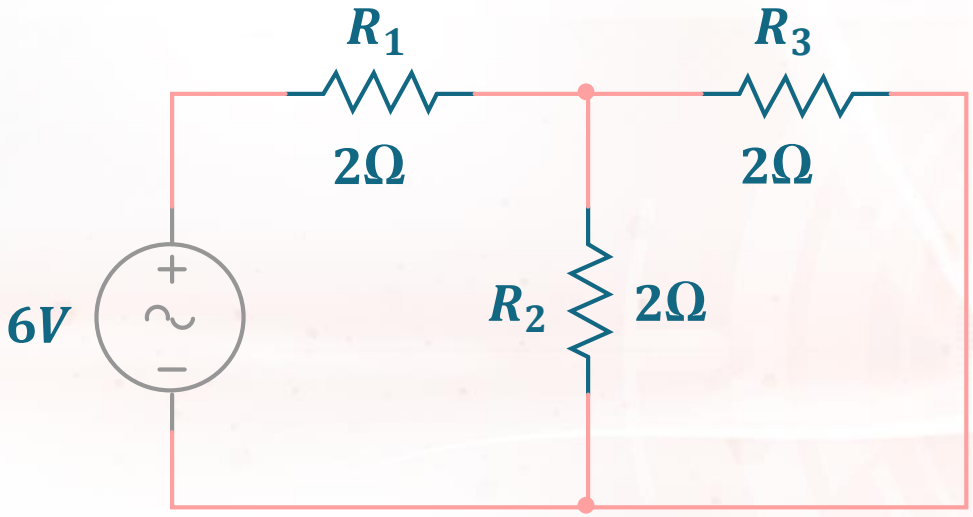
$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & \end{array} \\ & \begin{bmatrix} 2 & -4 & -2 & 12 \\ 0 & -10 & -16 & 48 \\ 0 & 0 & -4 & 12 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \end{array}$$

$$2a - 4b - 2c = 12$$

$$\frac{\cancel{2}a}{\cancel{2}} = \frac{6}{2}$$

$$a = 3$$

Determine the voltages v_1, v_2 and v_3 of the given network.



$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

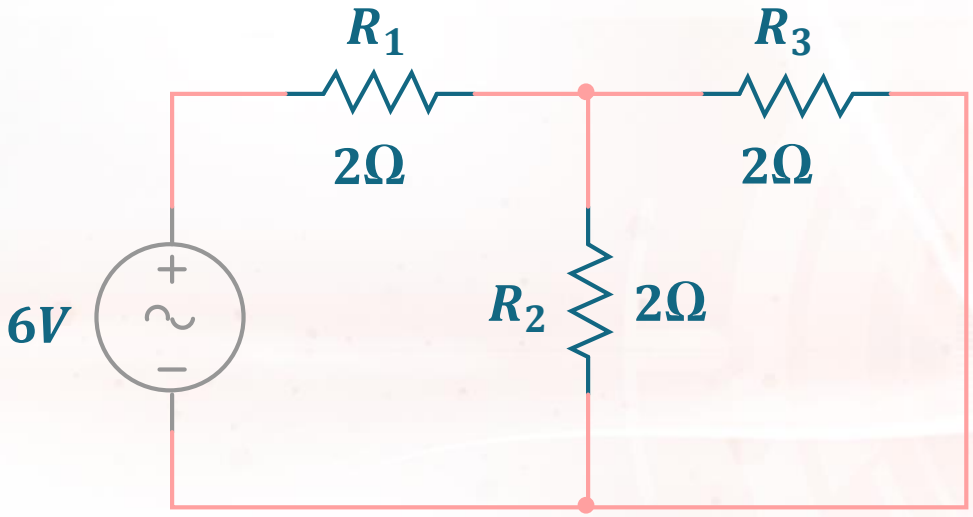
$$v_1 + v_2 + 0 = 6$$

$$0 - v_2 + v_3 = 0$$

Augmented matrix,

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 1 & 0 & 6 \\ 0 & -1 & 1 & 0 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Augmented matrix,

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 1 & 0 & 6 \\ 0 & -1 & 1 & 0 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

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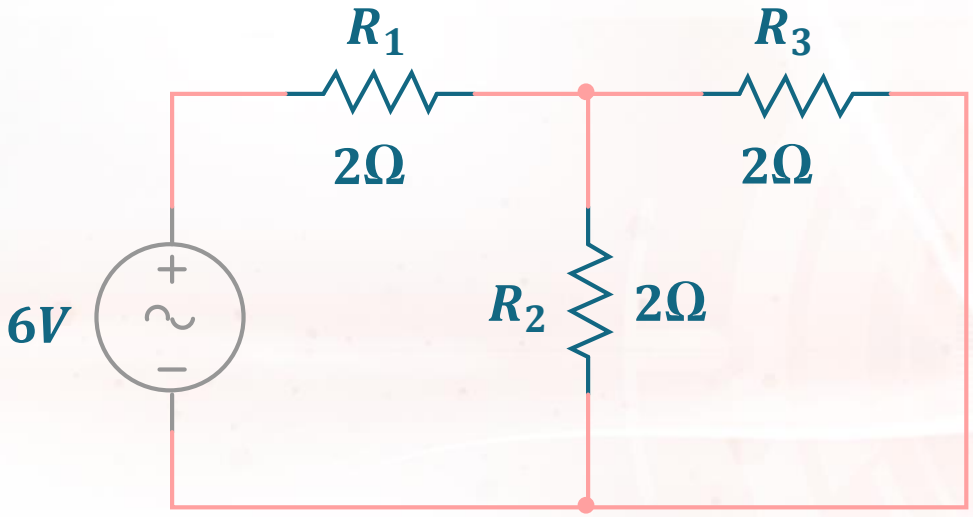

Augmented matrix,

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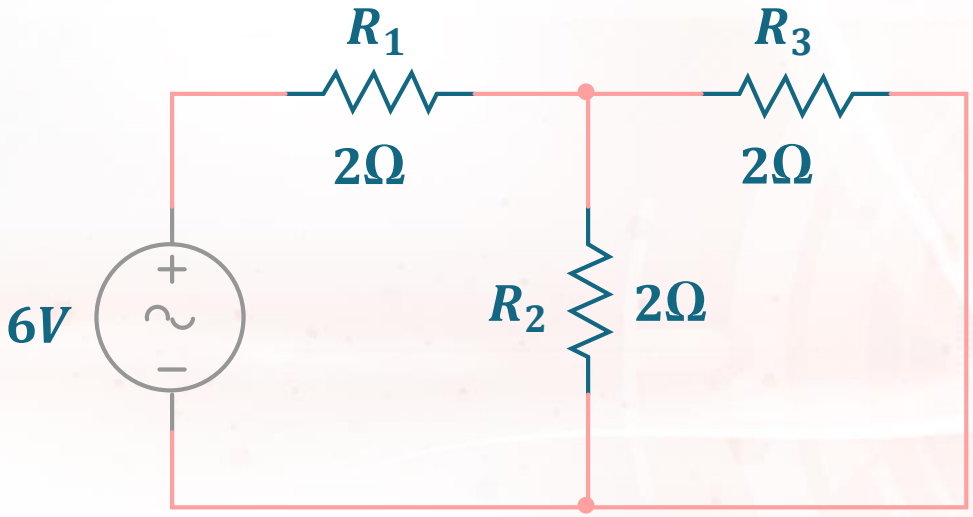


Augmented matrix,

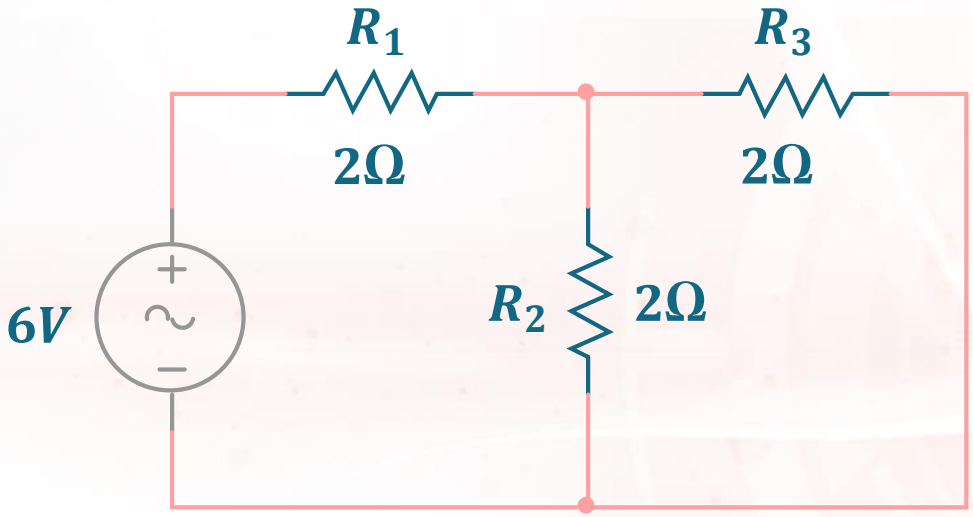
$$
\begin{array}{cc}
 & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\begin{array}{c} 2 \\ \\ \\ \end{array} & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 1 & 0 & 6 \\ 0 & -1 & 1 & 0 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}
\end{array}
$$

$-1 \quad 1 \quad 1$

$\rightarrow 2r_1 + r_2$

$$
\begin{array}{c}
\begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\
\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & -1 & 1 & 0 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array}
\end{array}
$$

Determine the voltages v_1, v_2 and v_3 of the given network.



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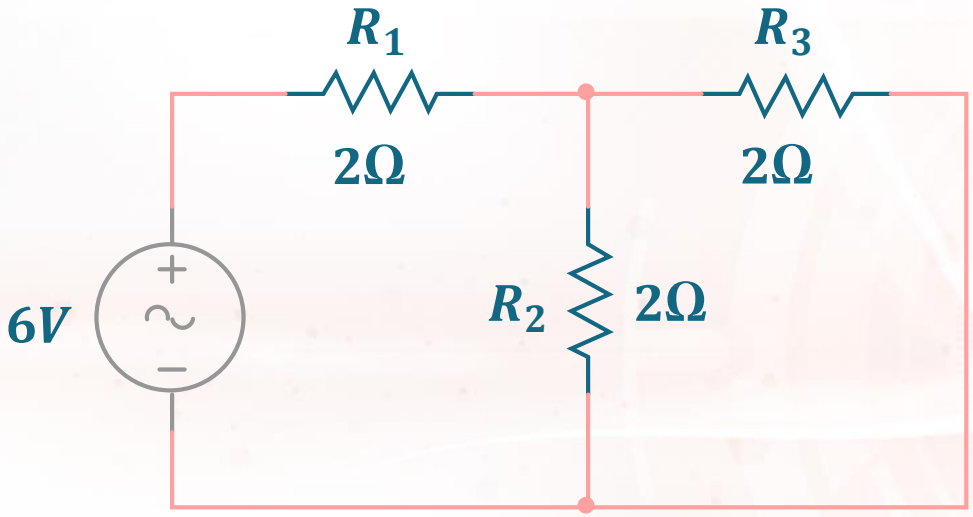
Determine the voltages v_1, v_2 and v_3 of the given network.



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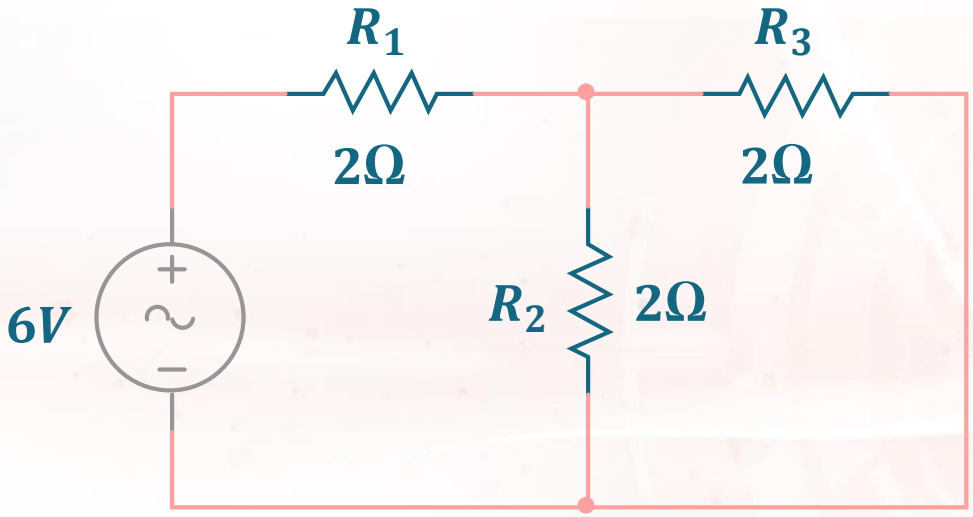
$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \begin{array}{l} r_1 \\ r_2 \\ r_3 \end{array} \end{array}$$

Determine the voltages v_1, v_2 and v_3 of the given network.



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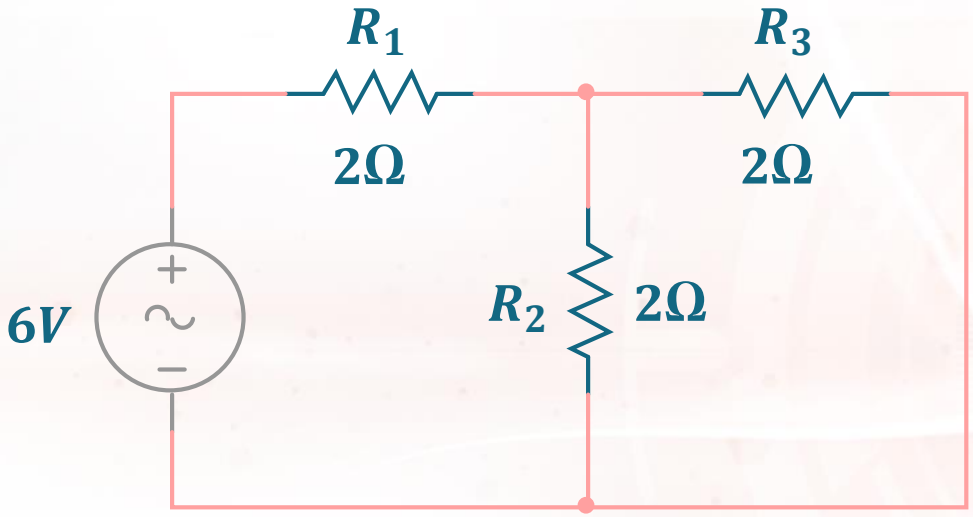
Determine the voltages v_1, v_2 and v_3 of the given network.



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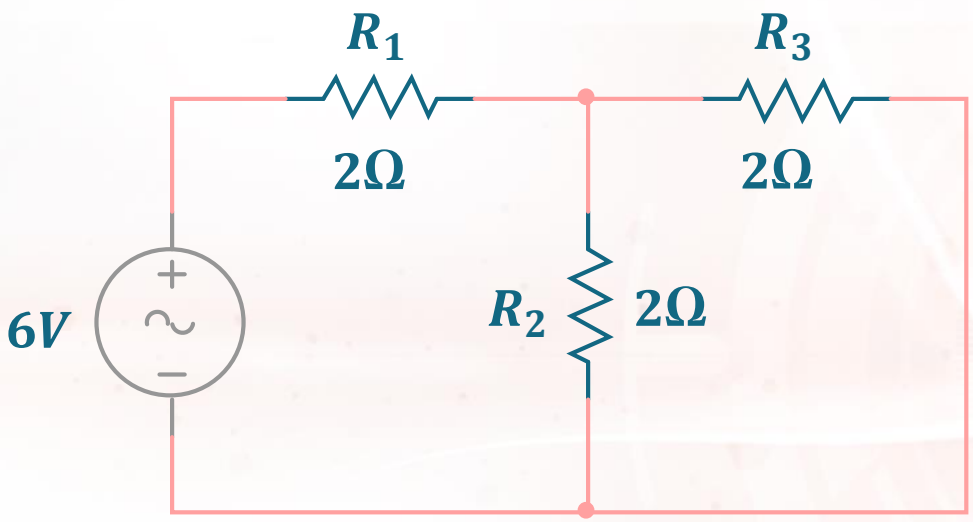
Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$$\boxed{v_3 = 2}$$

ans

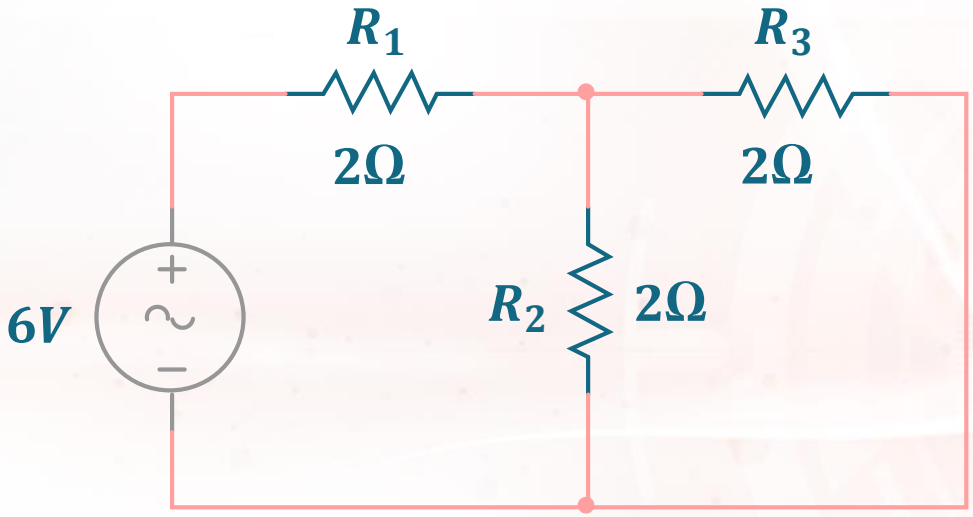
Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$\frac{\cancel{3}v_3 = 6}{\cancel{3}}$$

$$v_3 = 2$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Augmented matrix,

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

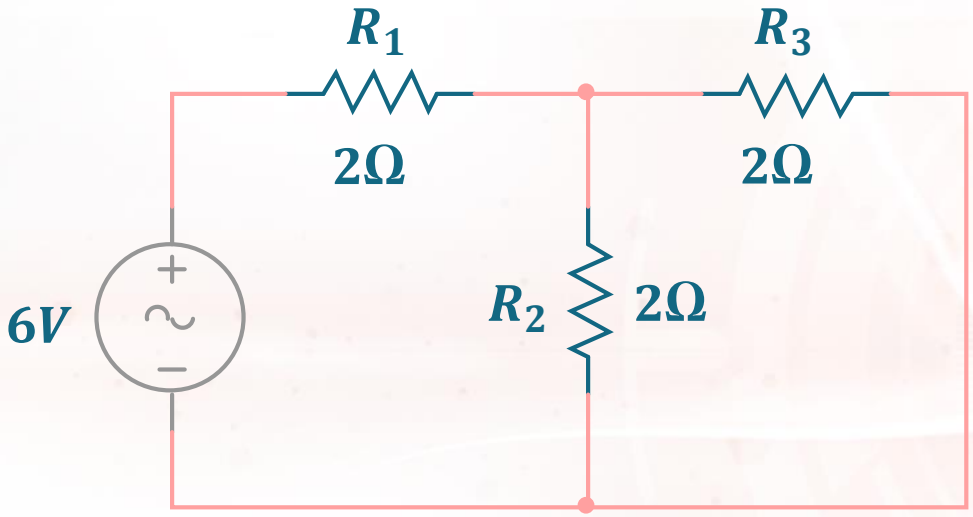
$$2v_2 + v_3 = 6$$

Solution,

$$\boxed{v_3 = 2}$$

ans

Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$$\boxed{v_3 = 2}$$

ans

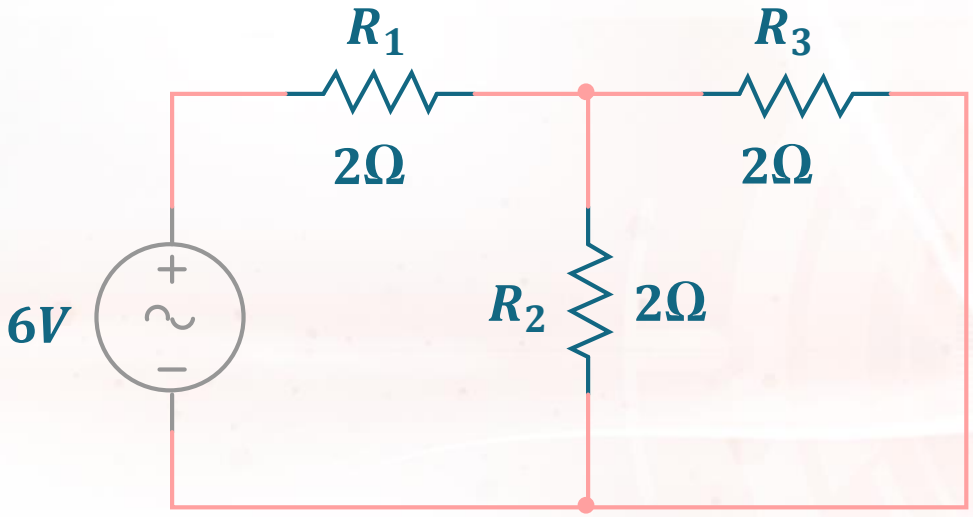
Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$2v_2 + v_3 = 6$$

$$2v_2 + 2 = 6$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$$\boxed{v_3 = 2}$$

ans

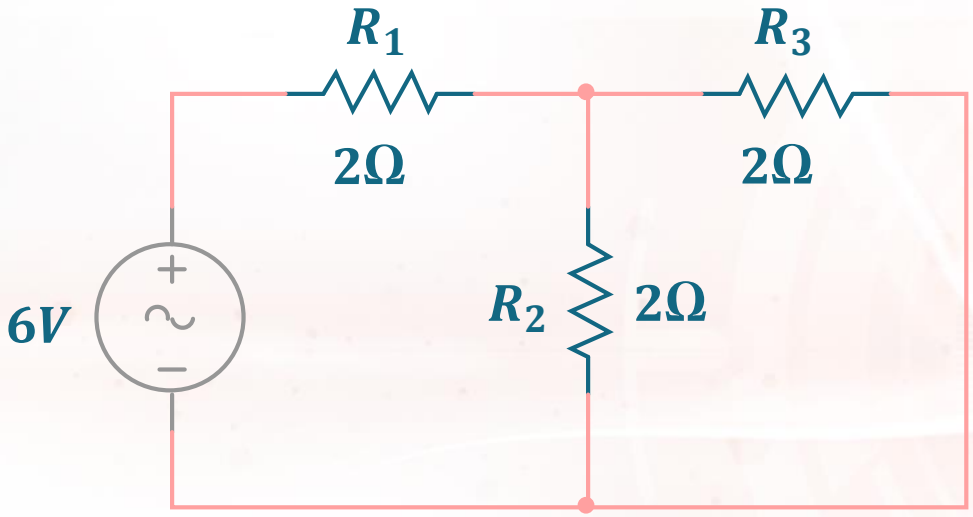
Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$2v_2 + v_3 = 6$$

$$2v_2 = 6 - 2$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$\boxed{v_3 = 2}$ *ans*

$\boxed{v_2 = 2}$ *ans*

Augmented matrix,

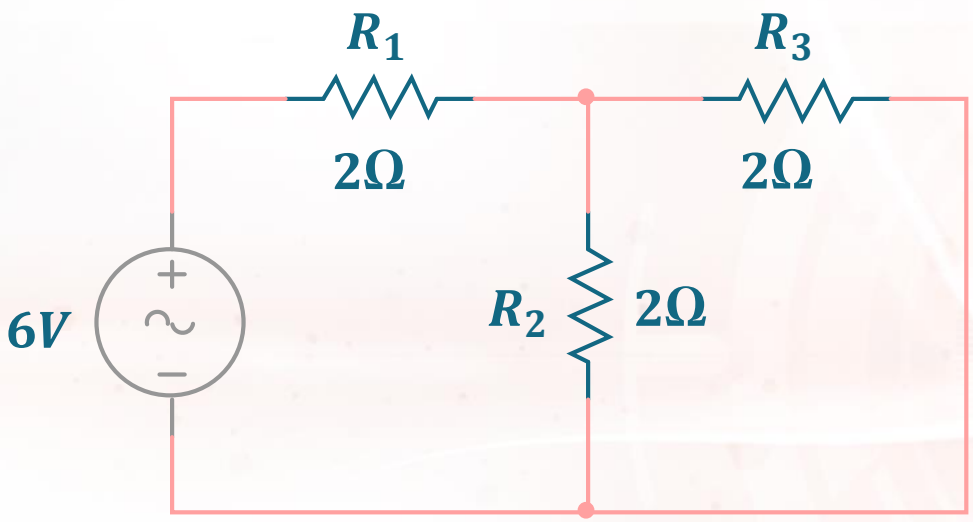
$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$2v_2 + v_3 = 6$$

$$\frac{\cancel{2}v_2}{\cancel{2}} = \frac{4}{2}$$

$$v_2 = 2$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Augmented matrix,

$$\begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

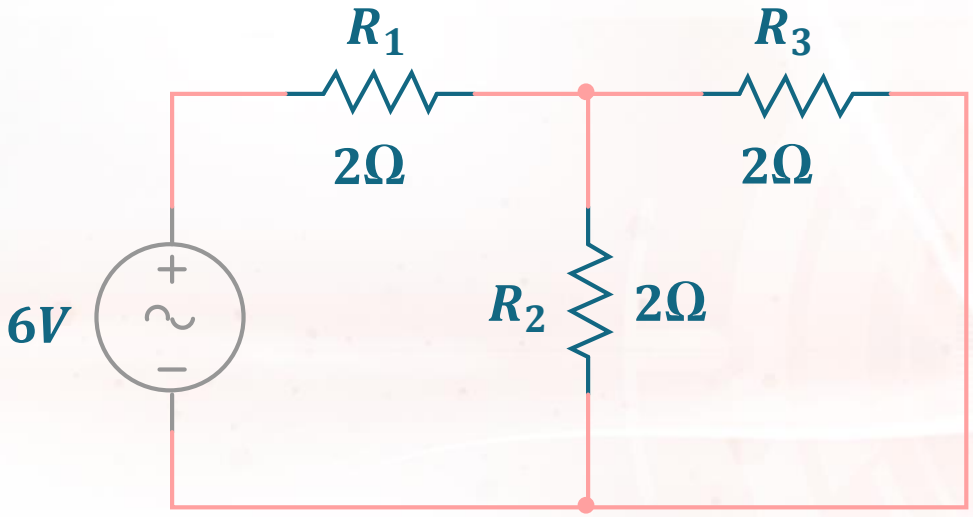
$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

Solution,

$\boxed{v_3 = 2}$ *ans*

$\boxed{v_2 = 2}$ *ans*

Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$$\boxed{v_3 = 2}$$

ans

$$\boxed{v_2 = 2}$$

ans

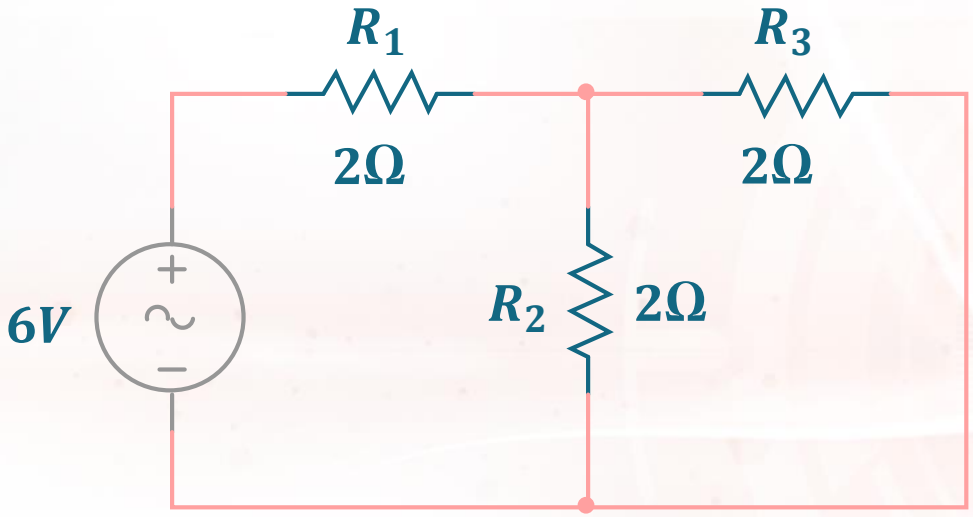
Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

$$-\frac{v_1}{2} + \frac{2}{2} + \frac{2}{2} = 0$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$$\boxed{v_3 = 2}$$

ans

$$\boxed{v_2 = 2}$$

ans

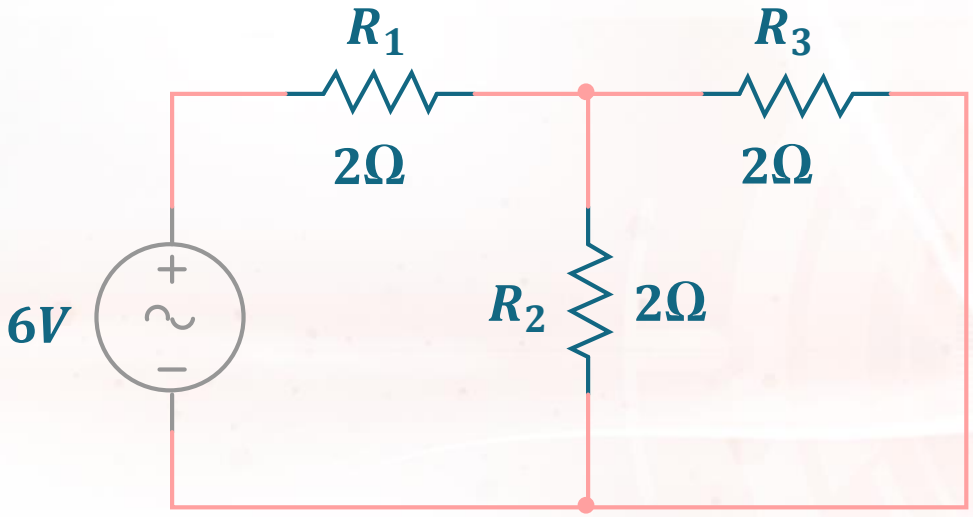
Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

$$-\frac{v_1}{2} + 1 + 1 = 0$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$\boxed{v_3 = 2}$ *ans*

$\boxed{v_2 = 2}$ *ans*

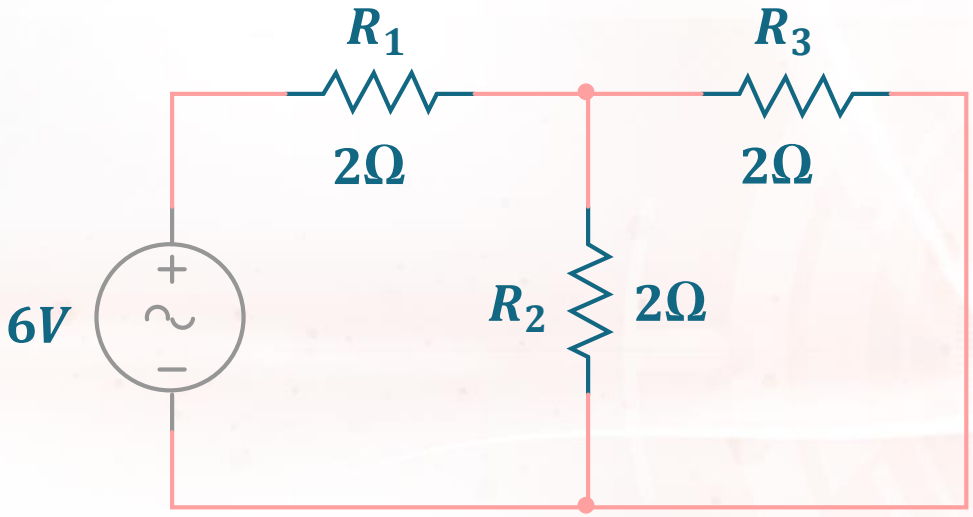
Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

$$-\frac{v_1}{2} \qquad + 2 = 0$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$\boxed{v_3 = 2}$ *ans*

$\boxed{v_2 = 2}$ *ans*

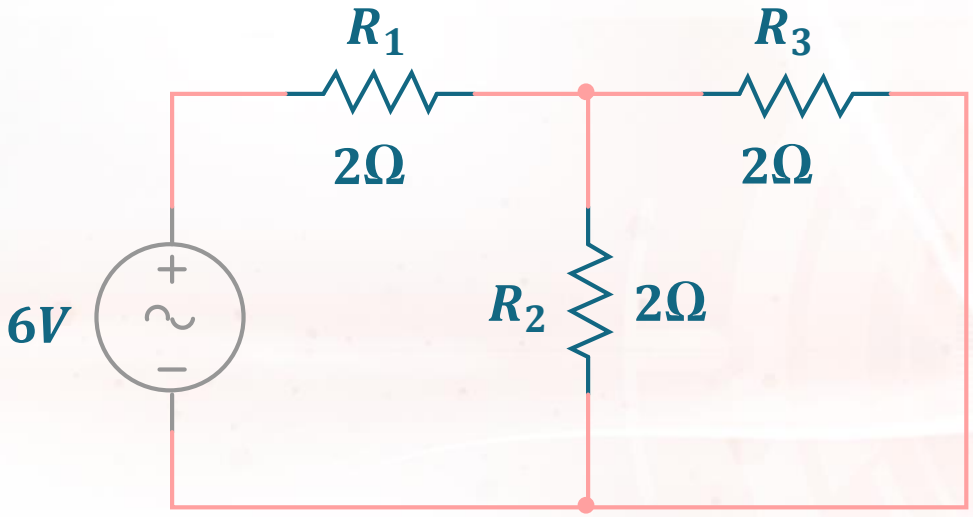
Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

$$-\frac{v_1}{2} = -2$$

Determine the voltages v_1, v_2 and v_3 of the given network.



Solution,

$\boxed{v_3 = 2}$ *ans*

$\boxed{v_2 = 2}$ *ans*

$\boxed{v_1 = 4}$ *ans*

Augmented matrix,

$$\begin{array}{cc} & \begin{array}{cccc} v_1 & v_2 & v_3 & \end{array} \\ & \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 2 & 1 & 6 \\ 0 & 0 & 3 & 6 \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \end{array}$$

$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

$$-v_1 = -2(2)$$

$$v_1 = 4$$

SEATWORK